

Simone Saitta, PhD

Assistant Professor in Cardiovascular Image Analysis

Amsterdam UMC & University of Amsterdam — qurAI Group (AI for quantitative healthcare) & CARA Lab

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PROFILE

I develop data-driven methods to make cardiovascular imaging more quantitative and objective. My work brings together deep learning, computational modeling, physics, and image analysis to support diagnosis, treatment planning, and image-guided cardiac procedures.

APPOINTMENTS

Assistant Professor in Cardiovascular Image Analysis

May 2024 – present

Amsterdam UMC, Department of Biomedical Engineering & Physics

- Manager of the CARA Lab (a joint initiative of Amsterdam UMC, Radboudumc and Abbott Vascular): development and clinical validation of artificial-intelligence systems that analyse intracoronary optical coherence tomography (OCT) — a catheter-based imaging technique — in real time during procedures in the cardiac catheterisation laboratory.
- Lead an independent research line on Marfan syndrome (a genetic disorder that weakens the aorta), combining medical imaging, biomechanics and machine learning to quantify a patient's risk of aortic-disease progression.
- Further work on deep learning for coronary plaque characterisation and detection of thin-cap fibroatheromas (rupture-prone plaques that cause heart attacks), and on geometric deep learning as a fast surrogate for blood-flow simulation. Visiting Researcher, Diagnostic Image Analysis Group (DIAG), Radboudumc.

Junior Assistant Professor / Postdoctoral Research Fellow

Nov 2023 – Apr 2024 / May – Nov 2023

2023

Politecnico di Milano, Milan, Italy

- Developed graph neural networks as fast surrogates for cardiovascular biomechanics, turning simulations that normally take hours into near-instant predictions.
- Co-supervised two doctoral candidates at Politecnico di Milano.

Doctoral Researcher

Nov 2019 – May 2023

Politecnico di Milano, Milan, Italy

- Developed deep neural networks for automatic 3D segmentation of medical images — computed tomography (CT), magnetic resonance imaging (MRI) and 3D transesophageal echocardiography.
- Integrated these networks into a mixed-reality system for neurosurgical guidance, published at MICCAI 2022; the work led to a filed patent.
- Reconstructed moving 3D blood-flow fields from MRI without labelled data (implicit neural representations), and estimated cerebral-aneurysm haemodynamics by combining imaging with physics (4D variational data assimilation).

Visiting Doctoral Researcher

Mar – Aug 2022

University of Cambridge, Dept. of Applied Mathematics and Theoretical Physics; Cambridge Image Analysis Group (host: Prof. Carola-Bibiane Schönlieb). Inverse problems in imaging: unsupervised denoising and super-resolution of flow-encoded MRI.

Earlier research roles

2018 – 2019

Research Engineer, National University of Singapore (4D image registration; host: Prof. C.H. Yap). Research Assistant, Imperial College London (non-invasive pressure estimation from MRI; computational fluid dynamics of aortic dissection; host: Prof. X.Y. Xu).

EDUCATION	PhD in Bioengineering, cum laude Politecnico di Milano, 2023 <i>Thesis: Data-driven approaches for quantification of complex biomarkers from medical imaging. External reviewers: Prof. Jelmer Wolterink (University of Twente), Prof. Sara Moccia (Scuola Superiore Sant'Anna).</i> MSc in Biomedical Engineering (Biomechanics & Biomaterials), cum laude Politecnico di Milano, 2018 <i>Thesis on non-invasive assessment of aortic coarctation from 4D-flow MRI. Final year at Imperial College London.</i> BSc in Biomedical Engineering Politecnico di Milano, 2015
AWARDS & GRANTS	- McKusick Fellowship Research Grant, The Marfan Foundation, 2023 — \$100,000, as Principal Investigator. Competitive international fellowship on machine learning for markers of aortic dissection in Marfan patients. <i>Editor's Choice</i> article, <i>Medical & Biological Engineering & Computing</i> (December 2024), for Riva, Saitta, et al., <i>left-ventricular diastolic vortex-ring characterisation in ischaemic cardiomyopathy</i>. - Switch2Product startup competition, 2022 — €30,000, for IPSE-XR, a deep-learning and mixed-reality solution supporting neurosurgery (core team member). - PhD awarded <i>cum laude</i>, Politecnico di Milano, 2023.
PATENTS	Two filed patents on AI-supported surgical guidance and vascular evaluation: <i>Method and system for the improvement of a virtual surgical procedure</i> — international (PCT) filing 2023; US national phase 2026 (App. 19/102,976). <i>System and method for the morpho-functional evaluation of the thoracic vascular district</i> — national filing, 2023.
SUPERVISION	Hired and supervised a postdoctoral researcher (Pascal Wijnjts) on neural surrogates for risk stratification in Marfan syndrome, and co-supervise nine doctoral candidates across Amsterdam UMC, Radboudumc and Politecnico di Milano: Ruben van der Waerden — AI-based assessment of intracoronary OCT (Radboudumc, 2023–27) Thijs Lutikholt — detecting rupture-prone plaque features from OCT (Radboudumc, 2023–27) Pierandrea Cancian — detection and correction of imaging artefacts in OCT (Amsterdam UMC, 2023–27) Joske van der Zande — OCT-guided coronary revascularisation (Radboudumc, 2023–27) Leah Heil — AI for 3D coronary-artery-tree modelling and OCT co-registration (Radboudumc, 2025–29) Xiaojin Gu — AI optimisation of coronary stent placement (Amsterdam UMC, 2024–28) Ioanna Gogou — non-invasive functional coronary assessment from CT angiography (Amsterdam UMC, 2025–29) Guido Nannini — deep learning for coronary-artery-disease risk assessment (Politecnico di Milano, 2021–25) Riccardo Munafò — deep learning for 3D echocardiography analysis (Politecnico di Milano, 2021–25) Also co-supervised six MSc theses (physics-informed neural networks for blood flow; statistical shape modelling of the aorta; mixed-reality neurosurgery; heart-valve imaging) and several BSc theses on deep-learning segmentation.
PUBLICATIONS	33 peer-reviewed publications — journal articles and papers at MICCAI, IEEE ISBI and SPIE Medical Imaging — with 751 citations (full list on Google Scholar, 2026). Recurring terms — OCT: intracoronary optical coherence tomography; CTA: coronary CT angiography; 4D-flow MRI: time-resolved 3D flow imaging; TAVI: transcatheter aortic-valve implantation; FFR: fractional flow reserve. 1. R. Volleberg, T. Lutikholt, . . . , S. Saitta , J. Thannhauser, N. van Royen. AI-based identification of rupture-prone coronary plaques (thin-cap fibroatheromas) and clinical outcomes: the PECTUS-AI study, <i>European Heart Journal</i> (2025).

2. R. Volleberg, R. van der Waerden, . . . , **S. Saitta**, et al. Deep-learning segmentation of the whole coronary vessel and automatic plaque quantification in OCT. *European Heart Journal – Digital Health* (2025).
3. R. Volleberg, D. Shin, **S. Saitta**, . . . , Z.A. Ali. Deep-learning measurement of plaque burden from OCT, validated against intravascular ultrasound. *JACC: Cardiovascular Interventions* (2025).
4. R.L.M. van Herten, . . . , **S. Saitta**, et al. World of Forms: deformable geometric templates for one-shot 3D surface reconstruction of coronary arteries from CT angiography. *Medical Image Analysis* (2025).
5. L. Heil, R. van der Waerden, R. Volleberg, J. Thannhauser, **S. Saitta**. Uncertainty analysis in intravascular OCT segmentation. *IEEE International Symposium on Biomedical Imaging (ISBI)* (2026, poster).
6. P. Cancian, **S. Saitta**, X. Gu, et al. Automatic detection and grading of attenuation artefacts in intracoronary OCT. *SPIE Medical Imaging* (2025).
7. G. Nannini, J. Suk, P. Rygiel, **S. Saitta**, et al. Predicting blood-flow fields directly on coronary-artery meshes: a benchmark of geometric deep-learning models. *Computers in Biology and Medicine* (2025).
8. R. Munafò, **S. Saitta**, . . . , E. Votta. Automatic segmentation of the moving mitral valve from 3D ultrasound using semi-supervised learning. *Medical & Biological Engineering & Computing* (2025).
9. **S. Saitta**, M. Carioni, S. Mukherjee, C.-B. Schönlieb, A. Redaelli. Implicit neural representations for unsupervised denoising and super-resolution of 4D-flow MRI. *Computer Methods and Programs in Biomedicine* (2024).
10. **S. Saitta**, F. Sturla, R. Gorla, . . . , A. Redaelli. A CT-based deep-learning system for automatic aortic-root anatomy assessment to plan transcatheter valve implantation (TAVI). *Computers in Biology and Medicine* (2023).
11. **S. Saitta**, L. Maga, C. Armour, . . . , et al. Data-driven generation of realistic 3D blood-flow inlet conditions for the aneurysmal aorta. *Computer Methods and Programs in Biomedicine* (2023).
12. M.C. Palumbo*, **S. Saitta***, . . . , et al. Mixed reality and deep learning for external ventricular drain placement in emergency neurosurgery (*equal contribution). *MICCAI* (2022).
13. **S. Saitta**, F. Sturla, A. Caimi, . . . , E. Votta, et al. A fully automated pipeline for thoracic-aorta anatomy assessment and endovascular-repair planning from CT. *Journal of Digital Imaging* (2022).

Invited book chapter (in preparation) — **S. Saitta** et al., in *Model Validation and Uncertainty Quantification in Biomechanics: From Soft Biological Tissue to Blood Flow*, Vol. 9 of Elsevier's *Biomechanics of Living Organs* series (J. Ohayon & Y. Payan, series editors).

INVITED TALKS

- Keynote — *Deep learning for image analysis in interventional cardiology*, COLIBRI Focus Workshop on Computational Medicine, University of Graz, 2025 (one of four invited keynote speakers).
- Invited talk — *Implicit neural representations for 4D-flow MRI*, International Congress on Industrial and Applied Mathematics (ICIAM), 2023.
- Invited talk — *Haemodynamic assessment in abdominal aortic aneurysm: imaging, simulation and data assimilation*, GIMC–SIMAI, 2022.
- Podium presentations at the European Society of Biomechanics (2023), ECCOMAS (2022) and the European Association of Cardiovascular Imaging (2021).

SERVICE

- Guest Editor, *Computers in Biology and Medicine* — special issue on cardiovascular digital twins (2025); co-authored the accompanying editorial.
- Organiser of a special session at the 19th Symposium on Computer Methods in Biomechanics and Biomedical Engineering (CMBBE), Vancouver, 2024.
- Invited external opponent at a doctoral half-time review, Karolinska Institutet (Sweden); regular reviewer of PhD and MSc theses and doctoral-defence committee member.
- Peer reviewer for *IEEE Transactions on Medical Imaging*, *Medical Image Analysis*, *Computers in Biology and Medicine*, *Computer Methods and Programs in Biomedicine*, and the MICCAI and IEEE ISBI conferences.

TEACHING &
SOFTWARE

Teaching Assistant, *Biomechanics* (third-year core course), Politecnico di Milano, 2020–24 — tutorials, group projects, and examinations. Course teaching at the University of Amsterdam / Amsterdam UMC begins in 2027.

[flow4D](#) — open-source Python toolkit for processing 4D-flow MRI and generating patient-specific inputs for blood-flow simulation; adopted internationally by cardiovascular-modelling groups. [INRs-4DFlowMRI](#) — reference implementation of implicit neural representations for 4D-flow MRI (published in *Computer Methods and Programs in Biomedicine*).

SKILLS

<i>Programming</i>	Python (expert), MATLAB, C; C++ (working knowledge)
<i>Machine learning</i>	PyTorch; geometric deep learning, implicit neural representations, self- and semi-supervised learning, generative models
<i>Medical imaging</i>	ITK/SimpleITK, VTK, elastix, 3D Slicer, DICOM pipelines; CT, MRI, 4D-flow MRI, 3D/4D echocardiography, intracoronary OCT
<i>Scientific computing</i>	FEniCS (physics-constrained optimisation, adjoint solvers), inverse problems, data assimilation
<i>Flow simulation</i>	Ansys, Star-CCM+, OpenFOAM, SimVascular, CRIMSON
<i>Infrastructure</i>	Docker, Singularity, GPU-accelerated high-performance computing, Git
<i>Languages</i>	Italian (native), English (full professional; TOEFL iBT 107)

REFERENCES

Prof. Alberto Redaelli (Politecnico di Milano) • Prof. Xiao Yun Xu (Imperial College London) • Prof. Selene Pirola (TU Delft). *Prof. Carola-Bibiane Schönlieb (University of Cambridge) available on request.* Full contact details on request.